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Course: Data Science

Exercise 2: SQL Aggregate functions & Operators

1 SELECT DISTINCT department
FROM enStudents;

Results:

department
IT
HR
Finance

2 SELECT department, AVG (age) AS avg-age
FROM Students
GROUP BY departments;

Results:

department	avg. ag
IT	20.5
HR	22
finance	23

3. SELECT department, COUNT (*) AS
Student - Count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

Results:

department	Student. Count
IT	2
HR	2

4. SELECT Student_id, name, age, department
FROM Students
WHERE age BETWEEN 21 AND 23;

Results:

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT Student_id, name, age, department
FROM Students
WHERE (department = 'IT' OR department = 'HR')
AND age > 21;

Results:

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q3 The courses Table

b. SELECT department, SUM(credits) As total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

Results:

department	total_credits
IT	11

7. SELECT Course_id, course_name, department, Credits
FROM courses
WHERE credits! = 4;

Results

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	finance	2
105	Statistics	HR	3

8. SELECT course_id, course_name, credits
FROM courses
ORDER By credits DESC
Limit 3;

Results

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

③. Enrollments table

9. SELECT
MAX (grade) As max_grade,
MIN (grade) As min_grade
AVG (grade) As Avg_grade
from enrollments

Results

Max grade	min_grade	avg_grade
90	78	84.6

10. SELECT course_id, Count (*) As enrollment_count
FROM enrollments
GROUP BY course_id;

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

④ Salaries table

11. SELECT
department
Sum (Salary) As total_salary,
Sum (bonus) As total_bonus
from salaries
Group By department;

Results:

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department, Avg (Salary) As avg_salary
FROM salaries
GROUP BY departments
HAVING Avg (Salary) > 55000;

results	department	avg Salary
	IT	61000
	finance	70000

13. SELECT

employee_id, name, Salary, bonus, (Salary + bonus) As
total_compensation

FROM Salaries

WHERE (Salary + bonus) > 60000;

results	employee_id	name	Salary	bonus	total_compensation
	1	Tom	60000	5000	65000
	3	Spike	70000	6000	76000
	4	Tyre	62000	5500	67500

⑤. Projects table

14. SELECT department,

Sum (budget) As total_budget, Avg (budget) As avg-budget

FROM projects

GROUP BY department

HAVING Avg (budget) > 70000;

results	department	total_budget	avg-budget
	IT	270000	135000
	finance	80000	80000

15. SELECT project_id, project_name, department, budget

FROM Projects

WHERE budget BETWEEN 50000 AND 120000

AND department != 'Marketing';

Results:

Project_id	Project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000